

hw5

Ch 11

13)

Take element $\frac{1}{2} + \mathbb{Z} = [\frac{1}{2}] \in \mathbb{Q}/\mathbb{Z}$, $[\frac{1}{2}]$ has order 2 because $[\frac{1}{2}] * [\frac{1}{2}] = [1] = \mathbb{Z}$ however no element in the rationals has order 2 because there is no elem in $\mathbb{Q}$ st $a + a = 0$ other than 0 which has order 1 so it is not isomorphic to $\mathbb{Q}$

14)

Canonical homomorphism: $\phi(g) = gN$

$\text{Ker}\phi = N$ because if $n \in n$, $\phi(n) = nN = N$

14. Let G be a finite group and N a normal subgroup of G . If H is a subgroup of G/N , prove that $\phi^{-1}(H)$ is a subgroup in G of order $|H| \cdot |N|$, where $\phi : G \rightarrow G/N$ is the canonical homomorphism.

Let H be a subgroup of G/N . Then $\text{Let } P = \phi^{-1}(H) = \{g \in G : \phi(g) \in H\}$

P is a group:

it has the identity elem e because H is a subgroup of G/N so N is in H and e is in N so e must be in P

It has inverses: Let $g \in P$ then $gN \in H \implies g^{-1}N \in H$ because H is a subgroup so $g^{-1} \in P$

Associative is obvious because $P \subseteq G$ and G is a group so P must be assoc

Order: Know that each $xN \in H$ has order $|N|$ and there are $|H|$ cosets total. Additionally each coset in H is disjoint so each coset has unique elements. Since P is all the unique elements in all of the cosets in H , then there are $|H| \cdot |N|$ elements in P .

15)

Let $\phi : G_1/H_1 \rightarrow G_2/H_2$ be def by $\phi(x) = \phi(x)$.

This is a homomorphism because $(xy) = \phi(xy) = \phi(x)\phi(y) = (x)(y)$

well def: let $x, y \in G_1/H_1$ where $x = y$. This means $xy^{-1} \in H_1$ and therefore

$\phi(xy^{-1}) = \phi(x)\phi(y^{-1}) \in H_2$ (we can do this because ϕ is a homomorphism and $\phi(H_1) \subseteq H_2$ is given). So then $\phi(x)\phi(y^{-1}) \in H_2 \implies \phi(x) = \phi(y)$ so well def

16)

16. If H and K are normal subgroups of G and $H \cap K = \{e\}$, prove that G is isomorphic to a subgroup of $G/H \times G/K$.

Let $\phi : G \rightarrow G/H \times G/K$ be def by $\phi(g) = (gH, gK)$

Well def: Let $x, y \in G$ where $x=y$. Then $\phi(x) = (xH, xK) = (yH, yK) = \phi(y)$

This is a homomorphism because :

$$\phi(xy) = ((xy)H, (xy)K) = (xHyH, xKyK) = (xH, xK)(yH, yK) = \phi(x)\phi(y)$$

$\text{Ker}(\phi) = \{g \in G : \phi(g) = (H, K)\}$. Let $x \in \text{Ker}(\phi)$. This means that we want $xH=H$ and $xK=K$ so $x \in H$ and $x \in K$ but the only element that satisfies that is e so $\text{ker}(\phi) = \{e\}$

By first isomorphism thm Ker is normal and $G/\text{Ker}(\phi) \cong \text{Im } \phi$. Since $\text{Ker} = \{e\}$, $G/\text{Ker}(\phi) \cong G$ and $\text{Im } \phi \cong G/H \times G/K$

So G is isomorphic to a subgroup of $G/H \times G/K$

Ch 13

1)

- Order 1: $\mathbb{Z}_1$
- Order 2: $\mathbb{Z}_2$
- Order 3: $\mathbb{Z}_3$
- Order 4: $\mathbb{Z}, \mathbb{Z}_2 \times \mathbb{Z}_2$
- Order 5: $\mathbb{Z}$
- Order 6: $\mathbb{Z}$
- Order 7: $\mathbb{Z}$
- Order 8: $\mathbb{Z}, \mathbb{Z} \times \mathbb{Z}_2, \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$
- Order 9: $\mathbb{Z}, \mathbb{Z}_3 \times \mathbb{Z}_3$
- Order 10: $\mathbb{Z}_1$
- Order 11: $\mathbb{Z}_{11}$
- Order 12: $\mathbb{Z}_{12}, \mathbb{Z} \times \mathbb{Z}_2$
- Order 13: $\mathbb{Z}_{13}$
- Order 14: $\mathbb{Z}_1$
- Order 15: $\mathbb{Z}_1$
- Order 16: $\mathbb{Z}_1, \mathbb{Z} \times \mathbb{Z}_2, \mathbb{Z} \times \mathbb{Z}, \mathbb{Z} \times \mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$
- Order 17: $\mathbb{Z}_1$
- Order 18: $\mathbb{Z}_1, \mathbb{Z} \times \mathbb{Z}_3$
- Order 19: $\mathbb{Z}_1$
- Order 20: $\mathbb{Z}_2, \mathbb{Z}_1 \times \mathbb{Z}_2$
- Order 21: $\mathbb{Z}_{21}$

- Order 22: $\mathbb{Z}_{22}$
- Order 23: $\mathbb{Z}_{23}$
- Order 24: $\mathbb{Z}_2, \mathbb{Z}_{12} \times \mathbb{Z}_2, \mathbb{Z} \times \mathbb{Z}_2 \times \mathbb{Z}_2$
- Order 25: $\mathbb{Z}_2, \mathbb{Z} \times \mathbb{Z}$
- Order 26: $\mathbb{Z}_2$
- Order 27: $\mathbb{Z}_2, \mathbb{Z} \times \mathbb{Z}_3, \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_3$
- Order 28: $\mathbb{Z}_2, \mathbb{Z}_1 \times \mathbb{Z}_2$
- Order 29: $\mathbb{Z}_2$
- Order 30: $\mathbb{Z}_3$
- Order 31: $\mathbb{Z}_{31}$
- Order 32: $\mathbb{Z}_{32}, \mathbb{Z}_1 \times \mathbb{Z}_2, \mathbb{Z} \times \mathbb{Z}, \mathbb{Z} \times \mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}_2, \mathbb{Z} \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$
- Order 33: $\mathbb{Z}_{33}$
- Order 34: $\mathbb{Z}_3$
- Order 35: $\mathbb{Z}_3$
- Order 36: $\mathbb{Z}_3, \mathbb{Z}_1 \times \mathbb{Z}_2, \mathbb{Z}_{12} \times \mathbb{Z}_3, \mathbb{Z} \times \mathbb{Z}$
- Order 37: $\mathbb{Z}_3$
- Order 38: $\mathbb{Z}_3$
- Order 39: $\mathbb{Z}_3$
- Order 40: $\mathbb{Z}, \mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{Z}_1 \times \mathbb{Z}_2 \times \mathbb{Z}_2$

2)

$$200 = 2^3 \cdot 5^2$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z} \times \mathbb{Z}$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}^2$$

$$\mathbb{Z}_{2^2} \times \mathbb{Z}_2 \times \mathbb{Z} \times \mathbb{Z}$$

$$\mathbb{Z}_{2^2} \times \mathbb{Z}_2 \times \mathbb{Z}^2$$

$$\mathbb{Z}_{2^3} \times \mathbb{Z} \times \mathbb{Z}$$

$$\mathbb{Z}_{2^3} \times \mathbb{Z}^2$$

4)

a)

$$\mathbb{Z}_{12} / 2 \cong \{\} \text{ because } \mathbb{Z}_{12} / 2 \cong \mathbb{Z}_2, \mathbb{Z}_{12} / 3 \cong \mathbb{Z}_3, \mathbb{Z}_{12} / 4 \cong \mathbb{Z}_2, \mathbb{Z}_{12} / 6 \cong \mathbb{Z}_2$$

$$\mathbb{Z}_{12} / 2 \cong \{\} \text{ because } \mathbb{Z}_{12} / 2 \cong \mathbb{Z}_2, \mathbb{Z}_{12} / 3 \cong \mathbb{Z}_3, \mathbb{Z}_{12} / 4 \cong \mathbb{Z}_2, \mathbb{Z}_{12} / 6 \cong \mathbb{Z}_2$$

$$\mathbb{Z}_{12} / 3 \cong \{\} \text{ because } \mathbb{Z}_{12} / 3 \cong \mathbb{Z}_3, \mathbb{Z}_{12} / 4 \cong \mathbb{Z}_2, \mathbb{Z}_{12} / 6 \cong \mathbb{Z}_2$$

b)

$$\mathbb{Z} / 2 \cong \{\} \text{ because } \mathbb{Z} / 2 \cong \mathbb{Z}_2, \mathbb{Z} / 3 \cong \mathbb{Z}_3, \mathbb{Z} / 4 \cong \mathbb{Z}_2, \mathbb{Z} / 6 \cong \mathbb{Z}_2, \mathbb{Z} / 12 \cong \mathbb{Z}_2$$

$\mathbb{Z}/2 \cong \mathbb{Z}_2$ because $\mathbb{Z}/2 \cong \mathbb{Z}_2$, $\mathbb{Z}/2 \cong \mathbb{Z}_2$, $\mathbb{Z}/2 \cong \mathbb{Z}_2$, $\mathbb{Z}/2 \cong \mathbb{Z}_2$

$\mathbb{Z}/3 \cong \mathbb{Z}_3$ because $\mathbb{Z}/3 \cong \mathbb{Z}_3$, $\mathbb{Z}/3 \cong \mathbb{Z}_3$, $\mathbb{Z}/3 \cong \mathbb{Z}_3$, $\mathbb{Z}/3 \cong \mathbb{Z}_3$

c)

Let $G = \langle r, s \mid r^k = 1, s^2 = 1, srs = r^{-1}s \rangle$

$G/\langle s \rangle \cong \mathbb{Z}_k$ because $G/\langle s \rangle \cong \mathbb{Z}_k$, $G/\langle s \rangle \cong \mathbb{Z}_k$, $G/\langle s \rangle \cong \mathbb{Z}_k$

$G/\langle r \rangle \cong \mathbb{Z}_2$ because $G/\langle r \rangle \cong \mathbb{Z}_2$, $G/\langle r \rangle \cong \mathbb{Z}_2$, $G/\langle r \rangle \cong \mathbb{Z}_2$

$G/\langle r^2, s \rangle \cong \mathbb{Z}_2$ because $G/\langle r^2, s \rangle \cong \mathbb{Z}_2$, $G/\langle r^2, s \rangle \cong \mathbb{Z}_2$, $G/\langle r^2, s \rangle \cong \mathbb{Z}_2$

d)

Let D_n be the dihedral group of order $2n$, where r is a rotation with $r^n = 1$ and s is a reflection with $s^2 = 1$.

$\langle r \rangle \cong \mathbb{Z}_n$ because $\langle r \rangle \cong \mathbb{Z}_n$, $\langle r \rangle \cong \mathbb{Z}_n$, $\langle r \rangle \cong \mathbb{Z}_n$

$\langle r^2 \rangle \cong \mathbb{Z}_{n/2}$ because $\langle r^2 \rangle \cong \mathbb{Z}_{n/2}$, $\langle r^2 \rangle \cong \mathbb{Z}_{n/2}$, $\langle r^2 \rangle \cong \mathbb{Z}_{n/2}$

e)

A_3 is the alternating group in 3 and e is the identity permutation.

$A_3 \times \mathbb{Z}_2 \cong A_3 \times \mathbb{Z}_2 \cong A_3 \times \mathbb{Z}_2$

because $(A_3 \times \mathbb{Z}_2)/\langle \mathbb{Z}_2 \rangle \cong A_3$, $(A_3 \times \mathbb{Z}_2)/\langle \mathbb{Z}_2 \rangle \cong A_3$, $(A_3 \times \mathbb{Z}_2)/\langle \mathbb{Z}_2 \rangle \cong A_3$

f)

$\mathbb{Z}_2 \times \mathbb{Z}_2 \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$

because $\mathbb{Z}_2 \times \mathbb{Z}_2 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, $\mathbb{Z}_2 \times \mathbb{Z}_2 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, $\mathbb{Z}_2 \times \mathbb{Z}_2 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$

g)

For n ,

$n \times n \cong n \times n$

because $n \times n \cong n \times n$ and n is simple for n , so $n \times n \cong n \times n$

h)

$\mathbb{Q}$ is infinitely divisible. For example, $\sqrt{2} \dots$ so it cant be broken into finite steps

7)

Let G be a torsion group so all elements of G has finite order and let G be finitely generated. Since all elements have a finite order, we know that G cannot be isomorphic to $\mathbb{Z} \times \dots \times \mathbb{Z}$ for any length > 0 because $\mathbb{Z}$ has a subgroup that is infinite. So then using this in conjunction with the fundamental theorem of finitely generated abelian groups,

$G \cong \mathbb{Z}_{p_1^{a_1}} \times \dots \times \mathbb{Z}_{p_n^{a_n}}$ and since $\mathbb{Z}_{p^a}$ are finite, G must be finite as well.

8)

8. Let $G, H,$ and K be finitely generated abelian groups. Show that if $G \times H \cong G \times K$, then $H \cong K$. Give a counterexample to show that this cannot be true in general.

Since G, H, K are finitely generated there is a unique decomposition such that

$$\begin{aligned} G &\cong \mathbb{Z}^n \times \mathbb{Z}_{p_{1,1}^{a_{1,1}}} \times \dots \times \mathbb{Z}_{p_{1,x}^{a_{1,x}}} \\ H &\cong \mathbb{Z}^m \times \mathbb{Z}_{p_{2,1}^{a_{2,1}}} \times \dots \times \mathbb{Z}_{p_{2,y}^{a_{2,y}}} \\ K &\cong \mathbb{Z}^k \times \mathbb{Z}_{p_{3,1}^{a_{3,1}}} \times \dots \times \mathbb{Z}_{p_{3,z}^{a_{3,z}}} \end{aligned}$$

$$\begin{aligned} \text{So } G \times H &\cong \mathbb{Z}^n \times \mathbb{Z}_{p_{1,1}^{a_{1,1}}} \times \dots \times \mathbb{Z}_{p_{1,x}^{a_{1,x}}} \times \mathbb{Z}^m \times \mathbb{Z}_{p_{2,1}^{a_{2,1}}} \times \dots \times \mathbb{Z}_{p_{2,y}^{a_{2,y}}} \\ &\cong \mathbb{Z}^{n+m} \times \mathbb{Z}_{p_{1,1}^{a_{1,1}}} \times \dots \times \mathbb{Z}_{p_{1,x}^{a_{1,x}}} \times \mathbb{Z}_{p_{2,1}^{a_{2,1}}} \times \dots \times \mathbb{Z}_{p_{2,y}^{a_{2,y}}} \end{aligned}$$

$$\text{Similarly } G \times K \cong \mathbb{Z}^{n+k} \times \mathbb{Z}_{p_{1,1}^{a_{1,1}}} \times \dots \times \mathbb{Z}_{p_{1,x}^{a_{1,x}}} \times \mathbb{Z}_{p_{3,1}^{a_{3,1}}} \times \dots \times \mathbb{Z}_{p_{3,y}^{a_{3,y}}}$$

Since they are isomorphic, the infinite and finite parts to line up. So

$$\mathbb{Z}^{n+m} \cong \mathbb{Z}^{n+k} \implies n + m = n + k \implies m = k \implies \mathbb{Z}^m \cong \mathbb{Z}^k$$

$$\begin{aligned} \text{Similarly } \mathbb{Z}_{p_{1,1}^{a_{1,1}}} \times \dots \times \mathbb{Z}_{p_{1,x}^{a_{1,x}}} \times \mathbb{Z}_{p_{2,1}^{a_{2,1}}} \times \dots \times \mathbb{Z}_{p_{2,y}^{a_{2,y}}} &\cong \mathbb{Z}_{p_{1,1}^{a_{1,1}}} \times \dots \times \mathbb{Z}_{p_{1,x}^{a_{1,x}}} \times \mathbb{Z}_{p_{3,1}^{a_{3,1}}} \times \dots \times \mathbb{Z}_{p_{3,y}^{a_{3,y}}} \\ \implies \mathbb{Z}_{p_{2,1}^{a_{2,1}}} \times \dots \times \mathbb{Z}_{p_{2,y}^{a_{2,y}}} &\cong \mathbb{Z}_{p_{3,1}^{a_{3,1}}} \times \dots \times \mathbb{Z}_{p_{3,y}^{a_{3,y}}} \text{ since each decomposition is unique} \end{aligned}$$

So we have

$$\begin{aligned} \mathbb{Z}^m &\cong \mathbb{Z}^k \text{ and } \mathbb{Z}_{p_{2,1}^{a_{2,1}}} \times \dots \times \mathbb{Z}_{p_{2,y}^{a_{2,y}}} \cong \mathbb{Z}_{p_{3,1}^{a_{3,1}}} \times \dots \times \mathbb{Z}_{p_{3,y}^{a_{3,y}}} \\ \implies \mathbb{Z}^m \times \mathbb{Z}_{p_{2,1}^{a_{2,1}}} \times \dots \times \mathbb{Z}_{p_{2,y}^{a_{2,y}}} &\cong \mathbb{Z}^k \times \mathbb{Z}_{p_{3,1}^{a_{3,1}}} \times \dots \times \mathbb{Z}_{p_{3,y}^{a_{3,y}}} \end{aligned}$$

So H and K are isomorphic

For counterexample, let $G = \mathbb{Z} \times \mathbb{Z} \times \dots$ infinitely, $H = \mathbb{Z}$ and $K = ,$

Then $G \times H \cong G \cong G \times K$ but H is not isomorphic to K

9)

Let G and H be solvable, then subnorm series for G (call it $\{G\}$) and H (call it $\{H\}$) Then

$$G_n \times H_m \quad G_{n-1} \times H_m \quad \dots \quad \{e\} \times H_m \quad \{e\} \times H_{m-1} \quad \dots \quad \{e\} \times \{e\}$$

Where G_n corresponds with G and H_m corresponds with H

Call this chain K

This is subnormal because any elem G_x in the solvable series of G is in K like so:

$\frac{G_x \times H_m}{G_{x-1} \times H_m} \cong G_x / G_{x-1} \times H_m / H_m$. This is abelian Because G is solvable and H_m / H_m is isomorphic to $\{e\}$ which is abelian so we have a cross product between two abelian groups which is abelian

Again any elem H_y in the solvable series of H is in Then for $\frac{e \times H_y}{e \times H_{y-1}} \cong e / e \times H_y / H_{y-1}$ which again is abelian because e / e is abelian and H is solvable

so we have shown for any k , K_k / K_{k-1} is abelian so K is solvable